

POWER BI

DASHBOARD & DATA ANALYTICS

Complete syllabus-based study material for students and internship learners

MODULE	Power BI fundamentals, visualization, data preparation, DAX, dashboards and publishing
FORMAT	Concept → Explanation → Step-by-step workflow → Example → Practice focus → Checkpoint
PROJECT	Superstore Report + interactive dashboard, publishing and sharing workflow

HOW TO STUDY

Follow the sequence. Build every visual yourself in Power BI Desktop, change fields and filters, and record what changed. The goal is to understand why a visual works—not only where to click.



DAX starter measures

```
Total Sales = SUM(Sales[Sales])
Total Profit = SUM(Sales[Profit])
Profit Margin = DIVIDE([Total Profit], [Total Sales])
```

Power BI Learning Path

Power BI is learned best as a workflow: connect to data, clean and transform it, model relationships, build calculations, create visuals, add interactions, validate the story, and publish/share the report.

Stage Student outcome

1. Foundations Understand BI, Power BI, visuals, report pages, fields, filters and the Power BI ecosystem.
2. Visuals Create and format common charts, cards, tables, matrices, slicers and advanced charts.
3. Modeling & filtering Understand relationships, hierarchies, filter context, drillthrough and interactive navigation.
4. Data preparation Use Power Query to connect, clean, transform and shape real-world data.
5. DAX Create measures and calculated logic for KPIs, comparisons, ratios and business questions.
6. Storytelling Use themes, conditional formatting, bookmarks, buttons and layout principles.
7. Service Publish, refresh, share, collaborate and manage dashboards in Power BI Service.
8. Project Build a Superstore report from raw data through an executive-ready dashboard.

ANALYST WORKFLOW Define the business question → collect/connect data → clean → model → calculate → visualize → explain → recommend. This follows the practical analyst workflow described in the supplied roadmap. I filecite I turn3file0 I L8-L25 I

Visual 1 • Category Sales

Tech

420

Office

310

Home

250

Visual 2 • Segment Mix

Consumer 46%

Corporate 34%

Home 20%

Visual 3 • Sales Trend

Q1 Q2 Q3 Q4

455 320

UNIT 1

Introduction to Power BI

Complete Overview of Power BI

Power BI is Microsoft's business intelligence and analytics platform for connecting to data, transforming it, modeling relationships, creating interactive reports and sharing insights. A typical solution includes Power BI Desktop for authoring, Power Query for data preparation, the data model for relationships/calculations, and Power BI Service for publishing and collaboration.

Desktop is where most report building happens.

Power Query prepares data.

The model organizes relationships and calculations.

The Service supports publishing, sharing, refresh and collaboration.

The Power of Data Visualization

Visualization turns rows of data into patterns that people can recognize quickly. A good visual answers a question, emphasizes the important comparison, and avoids unnecessary decoration. The chart type should follow the analytical question: compare categories, show trends, understand distributions, or show relationships.

Always define the question before choosing the visual.

Use titles, units, labels and context so the chart can be understood without narration.

What Are BI & BI Tools?

Business Intelligence (BI) is the practice of turning business data into information that supports decisions. BI tools help users connect data sources, transform data, analyze it and communicate results through reports and dashboards.

Examples of BI capabilities include reporting, dashboards, self-service analysis and scheduled refresh.

The tool is only part of BI—the business question and interpretation matter equally.

What Is Power BI?

Power BI is a BI platform used to build reports and dashboards from multiple data sources. It supports interactive filtering, drillthrough, calculations, data modeling and publishing.

Why Power BI?

Power BI is useful when an organization needs repeatable reporting instead of manually rebuilding spreadsheets. Once a model and report are designed, users can refresh data and interact with the same analytical structure.

Interactive exploration.

Reusable data models and measures.

Large ecosystem of connectors.

Integration with Microsoft products.

The Modern BI Choice

Modern BI emphasizes self-service exploration while retaining governed data models and shared reporting. Power BI supports both analyst-led report creation and broader business consumption.

Separate data preparation from presentation where possible.

Design reports for a specific audience and decision.

Power BI: Key Advantages

Key advantages include a broad connector ecosystem, Power Query for transformation, DAX for calculations, interactive visuals, strong Microsoft integration, publishing through Power BI Service, and extensibility through custom visuals and scripting.

Treat these as capabilities to practice, not just features to memorize.

UNIT 2

Creating Charts — Basic

Introduction to Basic Charts

Basic visuals are the building blocks of a Power BI report. Column and bar charts compare categories; pie and donut charts show part-to-whole relationships; funnel charts show stages; ribbon charts show changes in rank or composition.

Start with a clean field selection.

Use the visual's formatting pane to control labels, titles, axes and layout.

Create a Column Chart

A column chart compares numeric values across discrete categories. Add a categorical field to the X-axis and a numeric measure to the Y-axis. Sort by the business metric when ranking matters.

Example / Practice

Visualizations → Column chart → Axis: Category → Values: Sales → Format: Title, data labels, axis and sorting

Create a Stacked Column Chart

A stacked column chart compares totals while showing how subcategories contribute to each total. It is useful for sales by region split by category, for example.

Example / Practice

Axis: Region → Values: Sales → Legend: Category → Turn on data labels where readable

Create a Pie Chart

A pie chart divides a total into category slices. Use it when there are only a small number of meaningful categories and the part-to-whole relationship is the main question.

Avoid using pie charts for many categories.

Example / Practice

Legend: Category → Values: Sales → Format: Detail labels and title

Create a Donut Chart

A donut chart is a variation of a pie chart with a center hole. It is useful when a compact part-to-whole view is needed, but the same readability limitations apply.

Create a Ribbon Chart

A ribbon chart emphasizes how category ranking changes across an ordered dimension such as month. It helps answer which categories moved up or down in rank.

Example / Practice

Axis: Month → Legend: Category → Values: Sales

Create a Funnel Chart

A funnel visual represents sequential stages, such as leads → qualified leads → proposals → closed sales. The values normally decrease through the stages, making drop-off visible.

Example / Practice

Group: Stage → Values: Count of Leads

Include and Exclude

Power BI lets users focus a visual by including selected data or excluding selected categories. This is useful during exploration and can help investigate outliers or segments.

Use Include to keep the selected values.

Use Exclude to remove selected values.

Check whether the filter affects only the visual or the wider report context.

View Selected Data

View data lets you inspect the underlying or summarized data represented by a visual. It is useful for validating whether a visual is based on the values you expect.

Export in CSV

When permitted by the report and tenant settings, summarized or underlying data can be exported from a visual. Export permissions and available options depend on how the report is configured.

Treat exported data according to organizational privacy/security rules.

Overview / Recap

By the end of this unit, the student should be able to select an appropriate basic chart, map fields to the correct buckets, format it clearly, inspect selected data and understand visual-level filtering.

UNIT CHECKPOINT Complete a small practical exercise for UNIT 2: build the feature yourself, change at least one field/setting, and write 2–3 sentences explaining what changed and why.

Practice • Top Categories by Sales

Technology

520

Furniture

410

Office

290

UNIT 3

Tables and Matrix

Introduction to Table and Matrix

A table displays fields in rows and columns, while a matrix adds row/column grouping and hierarchical drill levels. Tables are useful for detailed records; matrices are useful for summaries across multiple dimensions.

Create a Table

Select the Table visual and drag fields into the Columns/Values area. Then format headers, grid, totals, column widths and data display.

Example / Practice

Table → Product, Region, Sales, Profit → Format → Grid, headers, values and totals

Formatting & Styling Tables

Good table formatting makes scanning easier. Use clear headers, appropriate number formats, alignment and controlled column widths. Avoid excessive borders or decorative formatting.

Conditional Formatting in Table

Conditional formatting highlights values based on rules, color scales, data bars or icons. It can quickly reveal high/low performance.

Example / Practice

Table → Format → Cell elements / Conditional formatting → Apply to Profit or Sales

Change Aggregations

Power BI may summarize numeric columns as Sum, Average, Minimum, Maximum or another aggregation. The correct aggregation depends on the business meaning of the field.

Sales usually uses Sum.

A rate may require Average or a calculated measure rather than a raw sum.

Identifiers should not normally be summed.

Create a Matrix

A matrix supports row and column groupings and is ideal for cross-tab analysis such as Sales by Region and Year. Add fields to Rows, Columns and Values.

Example / Practice

Rows: Region → Columns: Year → Values: Sales

Formatting & Styling Matrix

Format row headers, column headers, subtotals, grand totals, stepped layout, grid lines and indentation. Keep hierarchy depth readable.

Conditional Formatting in Matrix

Conditional formatting can be applied to matrix values to highlight performance patterns across rows and columns.

Hierarchies in Matrix

A hierarchy allows users to move from a higher-level category to more detailed levels, such as Year → Quarter → Month or Category → Subcategory → Product.

Build meaningful hierarchies in the model when they will be reused.

Use drill controls intentionally.

Total and Subtotal

Totals summarize the current visible context. Subtotals summarize intermediate groups. Always verify that the aggregation makes business sense.

Matrix vs Table

A table is best for flat detailed data. A matrix is best for grouped cross-tab summaries and hierarchical exploration.

Number Formatting

Number formatting controls how values appear: decimal places, thousands separators, currency, percentages and negative values. Formatting changes presentation, not the underlying value.

Use consistent units across a report.

Do not display a ratio as a raw decimal when the audience expects a percentage.

UNIT CHECKPOINT Complete a small practical exercise for UNIT 3: build the feature yourself, change at least one field/setting, and write 2–3 sentences explaining what changed and why.

Matrix Thinking • Sales by Region

Region Sales Margin

West I 420K 18%

East I 390K 15%

Central I 335K 11%

South I 280K 9% Illustrative matrix-style layout: use conditional formatting to highlight high/low performance.

UNIT 4

Cards and Filters

Cards and Filters

Cards show a small number of important metrics, while filters restrict what data is visible in visuals or pages. Together they create focused KPI sections and interactive reports.

Number Card

A number card displays one key metric such as Total Sales, Profit or Customer Count. Use a measure when the metric needs consistent business logic.

Example / Practice

Card → Data field: Total Sales measure → Format → Callout value, category label, title

Date Card

A date card displays a key date such as Last Refresh Date, Selected Date or Latest Order Date. A dedicated date measure is often useful for consistent logic.

Example / Practice

Last Refresh = MAX('Calendar'[Date])

Text Card

A text-style card can communicate a selected category, status or short business message. Use it for context rather than large amounts of prose.

Multi-Row Card

A multi-row card can show several fields or values together. It can be useful for a compact summary of multiple KPIs or attributes.

Format a Card

Format cards using consistent font sizes, labels, units, spacing and background. The most important metric should receive the strongest visual emphasis.

Filter on Visual

A visual-level filter affects only one visual. Use it when a chart needs a narrower data scope than the rest of the page.

Filter on This Page

A page-level filter affects visuals on the current report page. It is useful when every visual on the page should use the same selected segment or period.

Filter on All Pages

A report-level filter applies across the report. Use carefully because it changes context for multiple pages.

Drill Through

Drillthrough lets a user right-click a data point and navigate to a detail page filtered to that selected context. A good drillthrough page explains the selected entity in more depth.

Create a drillthrough field.

Design a detail page.

Add a Back button for navigation.

UNIT CHECKPOINT Complete a small practical exercise for UNIT 4: build the feature yourself, change at least one field/setting, and write 2–3 sentences explaining what changed and why.

UNIT 5

Slicers

Introduction to Slicers and Types

A slicer is an interactive filter placed directly on a report page. Slicers can work with text categories, dates, numbers and other fields depending on the visual configuration.

Text Slicer

Use a text/category slicer to let users select one or multiple categories such as Region, Category or Customer Segment. Format selection behavior and search for usability.

Example / Practice

Slicer → Field: Region → Format → Search/selection controls

Date Slicer

Date slicers allow users to filter by a date range, relative date or other date selection mode. They are especially useful for dashboards with time-based analysis.

Example / Practice

Slicer → Field: Order Date → Slicer type: Between / Before / After / Relative

Format Date Slicer

Configure date display, header, selection mode and layout. Make the selected time period obvious to the user.

Creating Multiple Slicers

Multiple slicers allow combined filtering, such as Region + Category + Year. Too many slicers can overwhelm the page, so prioritize the filters users actually need.

Keep related slicers aligned.

Consider a reset/clear-filter interaction when many slicers exist.

UNIT CHECKPOINT Complete a small practical exercise for UNIT 5: build the feature yourself, change at least one field/setting, and write 2–3 sentences explaining what changed and why.

Interactive Slicer Concept • Sales by Region

All East West Central South

Filtered Result • West Selected

Tech

210

Office

160

Home

95

UNIT 6

Advanced Charts

Introduction to Advanced Charts

Advanced charts help explain change, composition, relationships and distribution. The correct visual depends on the business question rather than the visual's novelty.

Waterfall Chart

A waterfall chart shows how positive and negative components move a starting value to an ending value. It is useful for profit bridges, budget variance and revenue-to-profit analysis.

Example / Practice

Category: Month/Component → Y-axis: Change Amount

Gauge Chart

A gauge compares a current value against a target or range. It is best for a small number of headline KPIs where progress toward a target matters.

Do not use gauges for detailed comparisons across many categories.

Line Chart

Line charts show trends over an ordered dimension, especially time. Multiple lines can compare segments, but too many lines reduce readability.

Example / Practice

X-axis: Month → Y-axis: Sales → Legend: Region

Stacked Area Chart

A stacked area chart shows how components contribute to a total over an ordered axis. It emphasizes volume and composition together.

Line vs Stacked Column

Combining a line and stacked column chart can show a total/volume measure alongside a trend or rate. Ensure the two axes have clear units and do not create misleading comparisons.

Line vs Clustered Column

A line plus clustered columns can compare two related measures or a category breakdown with an overall trend. Use secondary axes cautiously.

Scatter Plot

A scatter plot shows the relationship between two numerical measures. Bubble size, color or other encodings can add another dimension. It is useful for finding clusters and outliers.

Example / Practice

X-axis: Sales → Y-axis: Profit → Size: Quantity → Legend: Region

Treemap

A treemap represents hierarchical part-to-whole relationships using nested rectangles. Rectangle size represents the measure and color can represent another dimension.

Example / Practice

Group: Category → Details: Subcategory → Values: Sales

Line Chart Drill with Slicers

Combine a time-series line chart with slicers to let users explore the same trend by region, product, customer segment or date. The slicer changes filter context; the chart should respond automatically.

UNIT 7

Maps

Introduction to Maps

Maps visualize data geographically. Correct geographic categorization and clean location fields are critical because ambiguous place names can be mapped incorrectly.

Basic Map

Use a map visual with geographic location or latitude/longitude and a numeric measure such as Sales. Configure geographic categories carefully.

Example / Practice

Location: City/State → Size: Sales → Tooltips: Profit

Filled Map

A filled map shades geographic regions based on a measure. It is useful when the geography has meaningful boundaries such as states or countries.

Map with Pie Chart

A map with pie-style category composition can show both location and mix, where the visual supports that representation. Use it only when the added complexity improves the decision.

Format a Map

Map formatting includes zoom, labels, bubble size, data colors, tooltips and geographic category handling. Avoid decorative maps that do not add analytical meaning.

Validate locations.

Use a map only when geography is part of the question.

UNIT CHECKPOINT Complete a small practical exercise for UNIT 7: build the feature yourself, change at least one field/setting, and write 2–3 sentences explaining what changed and why.

Map-Inspired Regional View • Sales

North

340K

West

460K

Central

275K

East

410K

South

240K

UNIT 8

KPI & Other Functions

Introduction to Advanced Functions

Power BI includes analytical features beyond standard charts, including KPI visuals, Key Influencers, Decomposition Tree, scripting and custom/third-party visuals. These should be used when they answer a clear analytical question.

KPI Visualization Tool

A KPI visual compares a current indicator to a target and often displays a trend. Build the underlying measures first so the KPI logic is consistent.

Example / Practice

Indicator: Total Sales → Target: Sales Target → Trend axis: Date

Learn Updates in Power BI

Power BI changes frequently. Keep track of new visual capabilities, modeling features, Power Query improvements, DAX functionality and Service features. Always confirm current UI labels when teaching or documenting steps.

Key Influencers

Key Influencers analyzes which factors are associated with a selected outcome. It can help explore drivers of a metric, but the result should be interpreted as association rather than automatic causation.

Decomposition Tree

A decomposition tree lets users break down a measure by different dimensions interactively. It is useful for exploring why a total is high or low across categories.

Example / Practice

Analyze: Total Sales → Explain by: Region, Category, Segment, Product

Python & R Scripting

Power BI can integrate Python and R for specific analytical or visualization workflows where supported. Scripts require a correctly configured local environment and appropriate package versions.

Use scripting when it adds value beyond native Power BI capabilities.

Document dependencies so another user can reproduce the report.

Third-Party Extensions / Custom Visuals

Custom visuals extend Power BI beyond built-in chart types. Evaluate them for security, licensing, performance, accessibility and long-term maintenance before using them in professional reports.

Icons Introduction

Icons can support KPI status, navigation and conditional visual cues. Use consistent symbols and provide enough context that meaning is not dependent on color alone.

UNIT CHECKPOINT Complete a small practical exercise for UNIT 8: build the feature yourself, change at least one field/setting, and write 2–3 sentences explaining what changed and why.

UNIT 9

Insert & Action Functions

Introduction to Insert & Actions

Power BI's Insert tools help create report composition and navigation. Images, text, shapes and buttons can make a report feel like an application while still keeping the analytical content central.

Insert Image

Images can be used for logos, product visuals or explanatory graphics. Keep image quality and aspect ratio consistent.

Insert Text

Text boxes communicate titles, subtitles, definitions, instructions and insight statements. Good dashboard text tells the audience what to look for.

Insert Shapes

Shapes can create containers, separators, backgrounds and visual grouping. Use them to establish hierarchy rather than filling empty space unnecessarily.

Insert Buttons

Buttons provide interactive actions such as page navigation, bookmarks, drillthrough and web URLs. Consistent button placement improves usability.

Web URL Action

A button or visual action can open a web URL. This can connect a report to documentation, a product page or a supporting system when organizational policies allow it.

Page Navigation Action

Page navigation buttons let users move between report pages without relying on the page tabs. This supports a controlled dashboard experience.

Bookmark Action

Bookmarks capture a report state, including filters and visual visibility depending on configuration. They are useful for toggling views, creating navigation menus and storytelling sequences.

Drillthrough Action

A drillthrough action navigates to a detail page using the selected context. Pair it with a clear Back button and a page title that identifies the selected item.

Create Power BI Account

A Power BI account is used to access the Power BI Service and collaboration capabilities. Account availability, licensing and organizational policies affect which Service features are available.

UNIT CHECKPOINT Complete a small practical exercise for UNIT 9: build the feature yourself, change at least one field/setting, and write 2–3 sentences explaining what changed and why.

UNIT 10

Advanced Extra Charts

Introduction to Advanced Extra Charts

Custom and extended visuals can add specialized storytelling patterns. These should be used selectively: first prove that a standard visual cannot communicate the question clearly.

Scroller

A scroller visual can display a moving or scrolling sequence of values. It can be used for headlines or KPI strips, but should not replace clear static KPI cards when the values need to be scanned quickly.

Word Cloud

A word cloud represents frequency or importance of text terms. It is useful for exploratory text summaries, but exact comparisons are difficult, so use it as a supporting visual.

Infographics

Infographic-style visuals combine icons, numbers and short explanations. They can make a report approachable but should retain clear units and data provenance.

Drill-Down Donut Chart

A drill-down donut can move from a broad category to more detailed categories. The hierarchy should be logically meaningful and not contain too many levels.

Drill-Waterfall Chart

A drillable waterfall can let users move from a high-level variance explanation to more detailed contributors.

Play Axis Slicer

A play-axis style interaction can animate or step through a categorical/time dimension. Use it when change over time or category progression is the focus.

Animated Bar Chart Race

A bar chart race animates category rankings over time. It can be engaging for storytelling but should not replace a static chart when precise comparison is the main need.

Sunburst Chart

A sunburst displays hierarchical part-to-whole relationships as concentric rings. It is useful for nested categories but becomes difficult with deep hierarchies.

Sankey Chart

A Sankey visual shows flows between categories or stages, with band width representing magnitude. It is useful for movement such as source → channel → destination.

Calendar & Timeline Slicer

Calendar/timeline slicers help users select dates visually. Use them when the time period itself is an important part of exploration.

UNIT CHECKPOINT Complete a small practical exercise for UNIT 10: build the feature yourself, change at least one field/setting, and write 2–3 sentences explaining what changed and why.

UNIT 11

Create a Superstore Report

Create Reports & Dashboard Documentation

A professional report needs documentation: data sources, refresh assumptions, metric definitions, page purpose, filters and known limitations. Documentation makes the report maintainable and easier to review.

Collaborate, Share & Integrate Across Products

Power BI can be used with other Microsoft products and collaboration workflows. Sharing should respect permissions and organizational governance.

Get Power BI Samples

Microsoft and other trusted sources provide sample datasets and reports for learning. Samples are useful for practicing visual design and understanding model patterns.

Use Report Themes

Themes define consistent colors, fonts and visual defaults. A theme helps a multi-page report look like one product rather than a collection of unrelated charts.

Conditional Table Formatting

Use data bars, icons, color scales or rules to highlight important values. Keep formatting consistent with the business meaning.

Ways to Share Your Work

Reports can be shared through Power BI Service mechanisms depending on licensing and permissions. Choose the sharing method based on audience, security and whether recipients need to interact with the report.

Organize Work in Workspaces

Workspaces organize reports, semantic models and related content for teams. Use meaningful names and access permissions.

Publish to Web

Publish to web creates a public web-accessible version and should only be used for information that is intentionally public. Never use it for confidential or sensitive business data.

Superstore Dashboard Structure

A practical Superstore report can include an Executive Overview, Sales & Profit Analysis, Product/Category Analysis, Customer/Segment Analysis and a detailed drillthrough page. Keep each page focused on a decision.

Executive page: KPI cards + sales/profit trend + region/category comparison.

Analysis page: detailed trends and filters.

Detail page: product/customer-level information.

Navigation: buttons/bookmarks/drillthrough.

UNIT CHECKPOINT Complete a small practical exercise for UNIT 11: build the feature yourself, change at least one field/setting, and write 2–3 sentences explaining what changed and why.

UNIT 12

Dashboard Functions

Publish Report to Power BI Service

Publishing sends the report and its model from Desktop to a selected workspace in Power BI Service. After publishing, configure permissions, refresh and other Service settings as needed.

Example / Practice

Power BI Desktop → Home → Publish → Sign in → Select Workspace

Export Power BI Report to PPT, PDF & PBIX

Power BI supports export or download options depending on the report, permissions and Service/Desktop context. PBIX is the Power BI Desktop file; PDF/PPT are presentation-oriented outputs and may not preserve all interactivity.

Check whether your organization allows export.

Keep the PBIX source under controlled storage.

Pin a Live Dashboard

In Power BI Service, supported visuals can be pinned to dashboards. A live tile can retain interaction with the underlying report depending on how it is pinned/configured.

Comments, Share & Subscribe

Comments support collaboration; sharing grants access to authorized users; subscriptions can send report snapshots/notifications according to Service capabilities and permissions.

Automatic Refresh

Automatic refresh keeps data current according to the configured schedule and connection type. Refresh design depends on whether the data source is cloud-accessible, requires a gateway, and what the licensing/tenant supports.

Free Data Files for Practice

Practice datasets should contain realistic columns, multiple categories and enough records to create meaningful analysis. Superstore, sales, HR, customer and ecommerce datasets are common practice themes.

UNIT CHECKPOINT Complete a small practical exercise for UNIT 12: build the feature yourself, change at least one field/setting, and write 2–3 sentences explaining what changed and why.

Total Sales

\$1.25M

+8.4%

Profit

\$214K

+5.1%

Orders

3,842

+11.2%

Margin

17.1%

+0.9pp

UNIT 13

DAX Functions

DAX Overview

DAX (Data Analysis Expressions) is the formula language used in Power BI semantic models for measures, calculated columns and calculated tables. DAX is strongly tied to filter context and the data model.

Measures calculate results in the current filter context.

Calculated columns calculate row by row when the model is processed.

Use measures for dynamic analytical KPIs whenever possible.

Example / Practice

Total Sales = SUM(Sales[Sales])

Basic DAX Queries / Expressions

Begin with SUM, COUNT, DISTINCTCOUNT, AVERAGE, MIN, MAX and simple arithmetic. Then learn CALCULATE, which changes filter context and is central to analytical DAX.

Example / Practice

Total Sales = SUM(Sales[Sales]) Total Orders = DISTINCTCOUNT(Sales[Order ID]) Average Sales = AVERAGE(Sales[Sales])

Advanced DAX Queries

Advanced DAX includes CALCULATE with filter expressions, time intelligence patterns, iterator functions such as SUMX, conditional logic, variables, ranking and context manipulation.

Build a proper Date table for time intelligence.

Use DIVIDE rather than direct division when zero/blank denominators are possible.

Example / Practice

Profit Margin = DIVIDE([Total Profit], [Total Sales])

Sales LY = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date]))

DAX Use Cases

Typical use cases include year-over-year growth, running totals, profit margin, target achievement, ranking, share of total, moving averages and dynamic KPI calculations.

Example / Practice

YoY % = DIVIDE([Total Sales] - [Sales LY], [Sales LY])

DAX Notes and Datasets

Maintain a DAX notes section with the business meaning of each measure, required columns, expected result and test case. A measure is easier to maintain when its purpose is documented.

Name measures consistently.

Keep business logic in measures instead of repeating formulas across visuals.

Test measures under multiple slicer/filter contexts.

UNIT CHECKPOINT Complete a small practical exercise for UNIT 13: build the feature yourself, change at least one field/setting, and write 2–3 sentences explaining what changed and why.

UNIT 14

Power Query

Introduction to Power Query

Power Query is the data preparation and transformation layer in Power BI. It connects to sources and applies repeatable steps such as changing types, removing columns, splitting fields, merging tables and replacing values.

Think of Power Query as a reproducible data-cleaning pipeline.

Transform once, refresh repeatedly.

Data Sources in Power Query

Power Query can connect to files, folders, databases, web sources and many other connectors. The exact connector list depends on the Power BI environment and version.

Power Query Editor Interface

The editor includes the Queries pane, data preview, ribbon transformations, formula bar and Applied Steps. Each transformation is stored as a step that can be inspected and reordered where appropriate.

Data Cleaning Basics

Cleaning includes removing duplicates, correcting data types, trimming/cleaning text, handling nulls, replacing errors, renaming columns and validating values.

Example / Practice

Typical sequence: Source → Promote Headers → Change Types → Remove Errors → Trim/Clean Text → Remove Duplicates

Data Shaping

Shaping changes the structure of data so it can be analyzed efficiently. Common operations include filter rows, choose/remove columns, split/merge columns, pivot/unpivot, group by, merge queries and append queries.

Unpivot is especially useful when month columns need to become rows.

Merge is similar to a relational join.

Append stacks compatible tables vertically.

Common Real-World Use Cases

Power Query is useful for recurring files, monthly Excel reports, folder ingestion, cleaning inconsistent exports, combining regional files and preparing fact/dimension tables before modeling.

Design transformations to survive the next refresh.

Avoid manual edits in the source file when the transformation can be automated.

UNIT CHECKPOINT Complete a small practical exercise for UNIT 14: build the feature yourself, change at least one field/setting, and write 2–3 sentences explaining what changed and why.

Final Practical Project

Superstore Executive Dashboard

Project Brief

Build an interactive Superstore report from raw data to an executive-ready dashboard. The project should demonstrate the complete Power BI workflow: Power Query cleaning, data modeling, DAX measures, visuals, slicers, drillthrough, navigation, formatting, documentation and publishing.

Project area Required output

Data preparation Clean data types, missing values, duplicates and inconsistent text in Power Query.

Model Create appropriate relationships and a Date table where needed.

DAX Total Sales, Total Profit, Orders, Average Order Value, Profit Margin and at least two comparison measures.

Executive page KPI cards, sales/profit trend, category or region comparison, key insight text.

Analysis page Slicers, advanced charts, conditional formatting and detailed comparisons.

Drillthrough Create a detail page for a selected product/category/region.

Navigation Buttons/page navigation and at least one bookmark interaction.

Service Publish, configure refresh where applicable, and share only with authorized users.

Documentation Data sources, metric definitions, page guide, refresh notes and limitations.

Final Skill Checklist

Explain BI, Power BI Desktop, Power Query, semantic models and Power BI Service.

Build and format basic and advanced charts based on a business question.

Create tables, matrices, cards and slicers with correct aggregation and number formatting.

Apply visual, page and report filters and explain filter scope.

Use hierarchies, drillthrough, buttons, bookmarks and page navigation.

Build appropriate maps and validate geographic fields.

Use KPI, Key Influencers and Decomposition Tree appropriately.

Understand custom visuals, Python/R scripting and their practical limitations.

Prepare and shape data with Power Query.

Write basic and advanced DAX measures and explain filter context at a practical level.

Publish a report to Power BI Service and understand sharing, refresh and dashboard concepts.

Create a polished Superstore report and communicate at least five actionable insights.

FINAL RULE

Learn the feature → build it yourself → test with a different field → ask what business question it answers → explain the result.

The supplied roadmap also emphasizes building dashboards, data modeling, Power Query, DAX basics, filters and storytelling as core Power BI skills. I filecite I turn3file0 I L42-L48 I

Executive Trend • Monthly Sales

Jan Feb Mar Apr May

255 185

Profitability • Sales vs Profit

Illustrative Power BI visual Higher Higher

X: Sales Profit

Superstore Project Workflow

Use this as the practical sequence for the final project

1 Connect

Import Excel/CSV or another approved source.

2 Clean

Fix types, nulls, duplicates and text in Power Query.

3 Model

Create relationships and a useful Date table.

4 Calculate

Build DAX measures for KPIs and analysis.

5 Visualize

Create charts, slicers, drillthrough and navigation.

6 Publish

Publish, refresh, share and the document report.

Goal: turn raw data into a clear decision-ready dashboard, not just a collection of visuals.

Documentation Data sources, metric definitions, page guide, refresh notes and limitations.

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Power BI Visual Gallery

A dashboard-style example page

Use this page as a visual reference: KPI cards + trend + category comparison + composition.



Dashboard Insight Panel

What should the student notice?

- Sales trend is strongest in the final month.
- Technology is the largest category in this illustrative example.
- Consumer contributes the largest segment share.
- KPI cards summarize the most important measures.
- Slicers can change the dashboard context.

Illustrative values only — replace them with actual Superstore model measures.

Power BI student completion checklist

Use this page as the final self-assessment before completing the Superstore project.

- 01 Explain BI, Power BI Desktop, Power Query, semantic models and Power BI Service.
- 02 Build and format basic and advanced charts based on a business question.
- 03 Create tables, matrices, cards and slicers with correct aggregation and number formatting.
- 04 Apply visual, page and report filters and explain filter scope.
- 05 Use hierarchies, drillthrough, buttons, bookmarks and page navigation.
- 06 Build appropriate maps and validate geographic fields.
- 07 Use KPI, Key Influencers and Decomposition Tree appropriately.
- 08 Understand custom visuals, Python/R scripting and their practical limitations.
- 09 Prepare and shape data with Power Query.
- 10 Write basic and advanced DAX measures and explain filter context at a practical level.
- 11 Publish a report to Power BI Service and understand sharing, refresh and dashboard concepts.
- 12 Create a polished Superstore report and communicate at least five actionable insights.

FINAL RULE

Learn the feature → build it yourself → change a field → test the interaction

→ explain the business insight. A good Power BI report should answer a decision, not just look attractive.

POWER BI DASHBOARD BLUEPRINT

A complete executive-page structure

The source project specifies an Executive Overview with KPI cards, sales/profit trend, category or region comparison, filters, drillthrough and navigation. Use this as the build blueprint.



DAX measures behind the dashboard

```
Total Sales = SUM(Sales[Sales])
Total Profit = SUM(Sales[Profit])
Orders = DISTINCTCOUNT(Sales[Order ID])
Average Order Value = DIVIDE([Total Sales], [Orders])
Profit Margin = DIVIDE([Total Profit], [Total Sales])
```

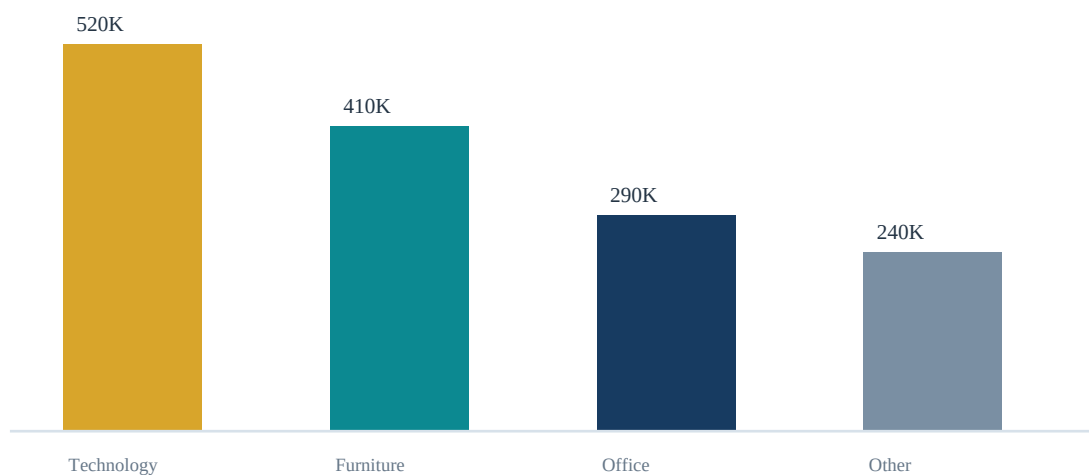
Practice: add Region and Category slicers and confirm that the KPI cards and charts respond.

BASIC CHARTS LAB

Compare, rank and explain category performance

Use bar/column charts for category comparisons, a line chart for ordered trends and a donut only when a small number of categories form a meaningful whole.

Category Sales — illustrative Superstore practice values



Monthly Sales Trend — illustrative values



Practice: replace the illustrative values with Sales from your Superstore model and sort categories descending.

DAX CODE LAB

Write measures that respond to filters

DAX practice set

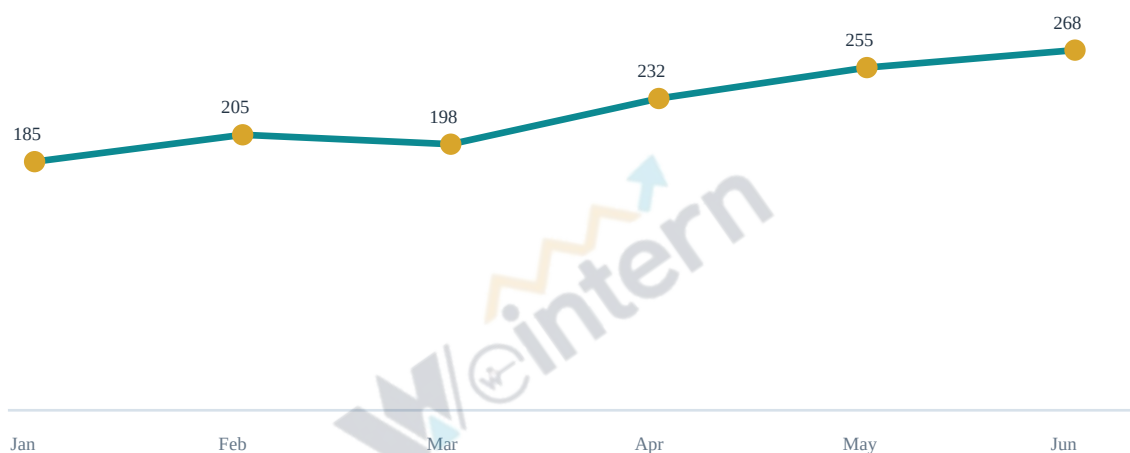
```
Total Sales = SUM(Sales[Sales])
Total Orders = DISTINCTCOUNT(Sales[Order ID])
Average Sales = AVERAGE(Sales[Sales])

Profit Margin =
DIVIDE([Total Profit], [Total Sales])

Sales LY =
CALCULATE(
    [Total Sales],
    SAMEPERIODLASTYEAR('Date'[Date])
)

YoY % =
DIVIDE([Total Sales] - [Sales LY], [Sales LY])
```

Monthly Sales Trend — illustrative values



Test the measures under Year, Region and Category filters. Measures should change with filter context.

POWER QUERY CODE LAB

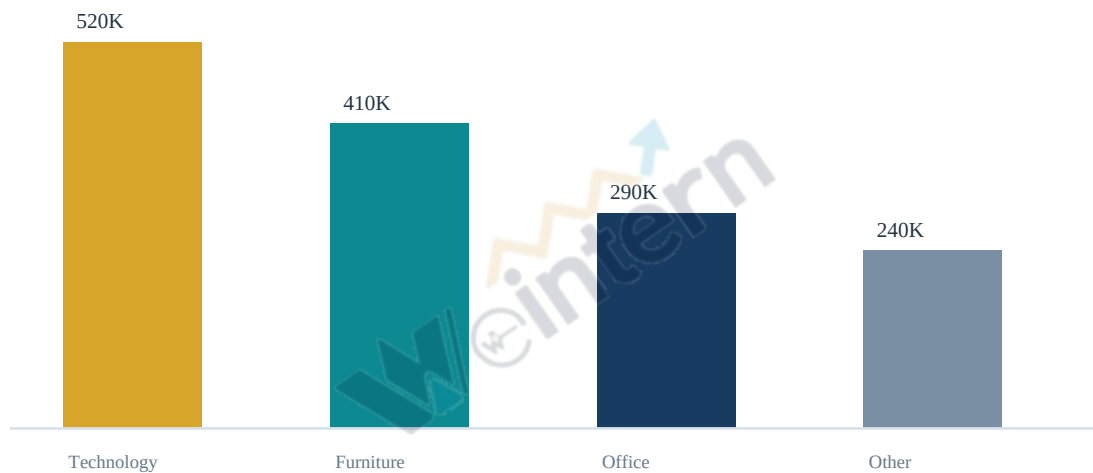
Build a repeatable cleaning pipeline

The source material describes Power Query as a reproducible pipeline. The example below demonstrates the same sequence: connect → promote headers → types → clean text → remove errors → remove duplicates.

Power Query M example

```
let
    Source = Excel.Workbook(File.Contents("Superstore.xlsx"), null, true),
    Orders = Source[[Item="Orders",Kind="Sheet"]][Data],
    Headers = Table.PromoteHeaders(Orders, [PromoteAllScalars=true]),
    Types = Table.TransformColumnTypes(Headers,
        {"Order Date", type date},
        {"Sales", type number},
        {"Profit", type number})),
    CleanText = Table.TransformColumns(
        Types, {"Category", Text.Trim, type text}),
    NoErrors = Table.RemoveRowsWithErrors(CleanText),
    NoDuplicates = Table.Distinct(NoErrors)
in
    NoDuplicates
```

Category Sales — illustrative Superstore practice values



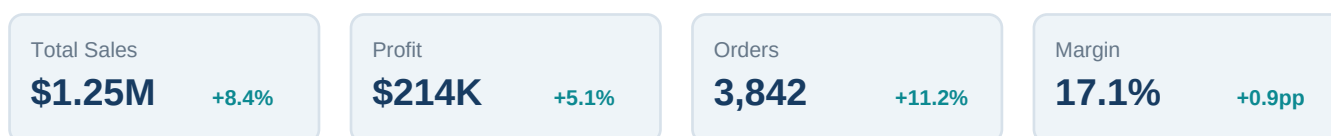
Practice: refresh the query after adding another source file and verify the same transformations repeat.

FILTERS, SLICERS & DRILLTHROUGH

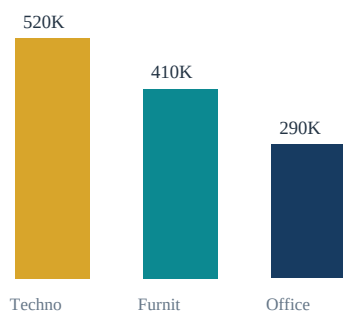
Make the dashboard interactive

Slicers can filter by Region, Category, Customer Segment or Order Date. Visual-level filters affect one visual; page-level filters affect the current page; report-level filters affect the report.

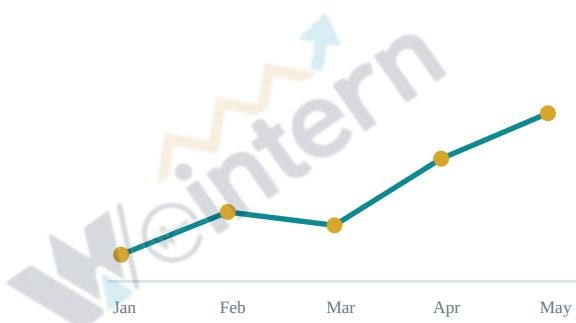
Interaction	Use it for	Student test
Slicer	Interactive selection	Choose West; verify every connected visual
Visual filter	Narrow one chart	Show only Technology
Page filter	Common page context	Keep one Region across the page
Drillthrough	Detail investigation	Right-click a category and open detail
Bookmark	Saved report state	Create an alternate view



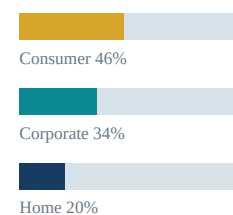
Sales by Category



Monthly Sales Trend



Segment Mix



Practice: test the same report with a single Region slicer, then reset it and compare the insight.

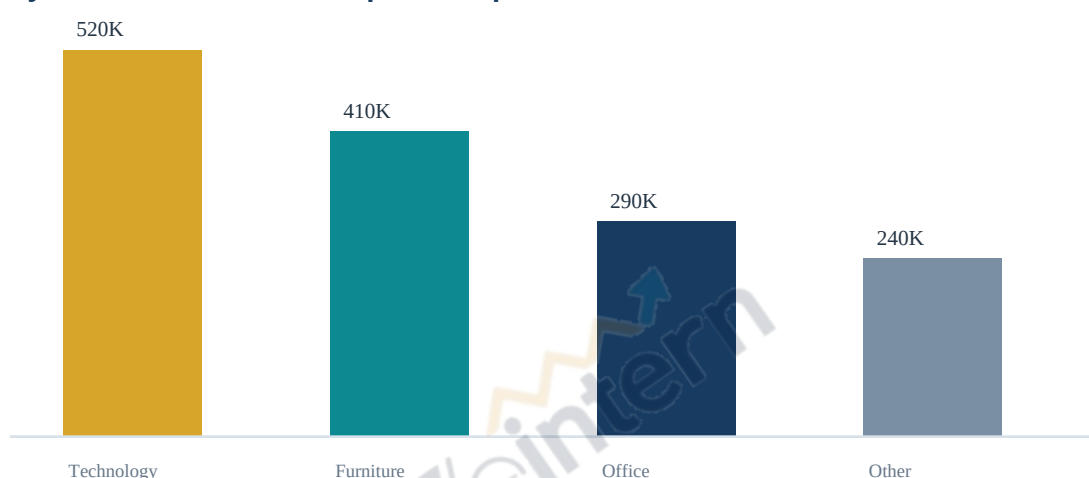
ADVANCED VISUALS LAB

Match the visual to the analytical question

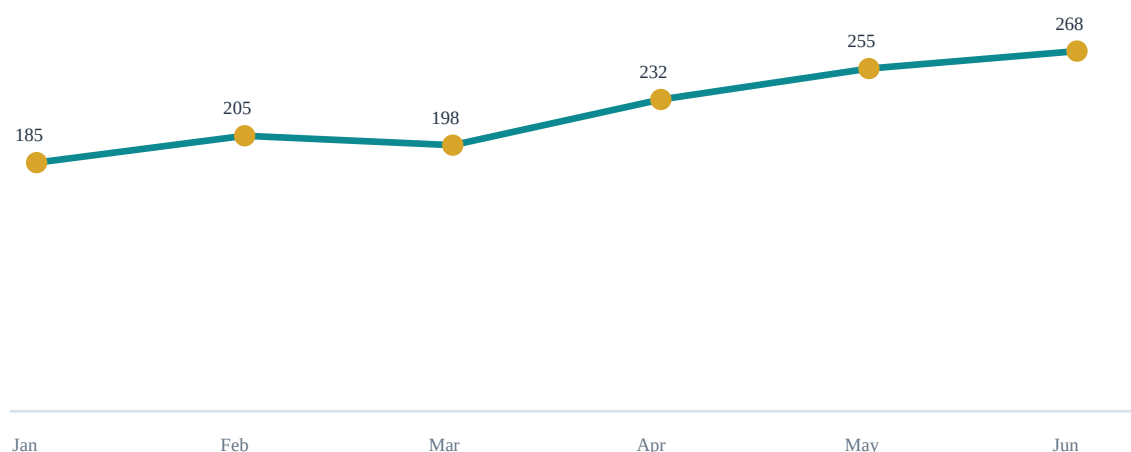
The source material covers waterfall, gauge, line, stacked area, combo charts, scatter and treemap. Use them when they clarify a business question.

Question	Visual	Typical Power BI fields
What changed the result?	Waterfall	Category/Component + Change
How are two measures related?	Scatter	Sales + Profit + optional Size
How is a hierarchy composed?	Treemap	Category → Subcategory + Sales
How does a KPI compare with target?	Gauge/KPI	Indicator + Target + Trend

Category Sales — illustrative Superstore practice values



Monthly Sales Trend — illustrative values



POWER BI SERVICE & PUBLISHING

From Desktop report to controlled sharing

The source material covers publishing, workspaces, refresh, sharing, comments, subscriptions and export. Use this checklist before sharing a report.

Step	Action
1	Power BI Desktop → Sign in → Publish
2	Select the correct workspace
3	Validate report pages, filters and measures
4	Configure permissions for authorized users
5	Configure refresh where applicable
6	Document data sources, metrics and limitations
7	Test the shared report from the audience's perspective

Service checklist

Publish workflow

Desktop → Sign in → Publish → Workspace → Permissions → Refresh → Share



SUPERSTORE PROJECT — FINAL BUILD

Turn the syllabus into one decision-ready dashboard

Required outputs include cleaned data, relationships and a Date table, DAX measures, an Executive page, an Analysis page, drillthrough, navigation, publishing and documentation.



Final skill check

✓ Explain Power BI Desktop, Power Query, semantic models and Service. ✓ Build charts from a business question. ✓ Create measures and explain filter context. ✓ Use slicers, drillthrough, buttons and bookmarks. ✓ Publish and document the report. ✓ Communicate at least five actionable insights.

Final rule: Learn the feature → build it yourself → change a field → test the interaction → explain the business insight.

POWER BI INTERVIEW & PRACTICE QUICK REFERENCE

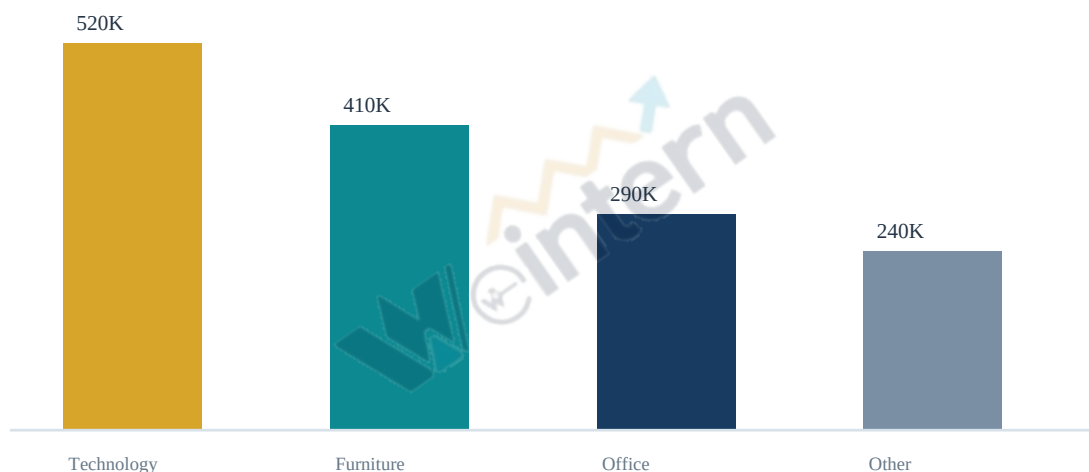
Questions a student should be able to answer after the syllabus

Question	What to explain
Power BI Desktop vs Service	Desktop is for authoring; Service supports publishing, sharing, refresh and collaboration.
Power Query vs DAX	Power Query prepares/shapes data; DAX calculates analytical logic in the model.
Measure vs calculated column	Measures respond to filter context; calculated columns are evaluated row by row during model processing.
Why a Date table?	A proper date dimension supports consistent time intelligence and date-based analysis.
Why use DIVIDE?	It safely handles zero/blank denominators for ratios such as profit margin.
What makes a good dashboard?	A clear business question, readable visuals, meaningful filters and an insight/action.

Core DAX to remember

```
Total Sales = SUM(Sales[Sales])
Total Profit = SUM(Sales[Profit])
Profit Margin = DIVIDE([Total Profit], [Total Sales])
Orders = DISTINCTCOUNT(Sales[Order ID])
```

Category Sales — illustrative Superstore practice values



Practice challenge: build a one-page Superstore dashboard, explain three insights, and defend why each visual answers a business question.